

Elvis Ramirez

Plano, TX, United States (open to relocation)

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[GitHub](#), eramirez.dev

Software engineer with a BS in Computer Science from the University of Texas at Dallas (August 2025), focused on backend and infrastructure systems that stay correct under failure. Built and operate a payments ledger with ACID and idempotency guarantees, a Go rate limiter sustaining 8,000+ req/s at p99 under 5 ms, an LLM agent orchestration platform on a durable Postgres-backed queue, and a self-built 9-host bare-metal Kubernetes platform delivered through Terraform and ArgoCD GitOps. Drives failures to root cause, documents them as runbooks, and makes engineering decisions from measured trade-offs. Uses Claude Code and GitHub Copilot in a daily workflow.

Core Competencies

- **Languages:** Go, Python, Java, SQL, Bash; C; TypeScript and Rust (actively ramping).
- **Backend & Distributed Systems:** REST API design, microservices, Kafka event streaming, PostgreSQL, Redis, Cassandra, idempotency, ACID transactions, durable queues, fault tolerance.
- **Infrastructure & Cloud:** Kubernetes (k3s), Docker, Terraform, ArgoCD GitOps, Ansible, GitHub Actions CI/CD, AWS, VMware ESXi 8, Linux, bare-metal servers with iDRAC/IPMI.
- **AI & Agentic Systems:** Multi-agent orchestration, self-hosted LLM inference (llama.cpp), schema-constrained outputs (Pydantic), output verification against source-of-truth data, human-in-the-loop review gates; Claude Code and GitHub Copilot in daily workflow.
- **Observability & Reliability:** Prometheus, Grafana, structured logging, alerting, performance profiling, incident response, root-cause analysis, runbooks.
- **Networking & Security:** VLAN and switch fabric design, DNS/DHCP, OAuth2/OIDC, HashiCorp Vault (PKI, secrets), TLS, least-privilege access design.

Technical Projects

- **Python, FastAPI, PostgreSQL, Redis, Kafka** github.com/BetV3/payments-api
Payments API & Ledger System
 - Owned a financial backend from design through deployment.
 - Implemented a double-entry ledger with ACID guarantees where no failure path, including client retries and partial failures, can duplicate or lose money movement.
 - Built 10 REST endpoints with idempotency keys and OAuth2/OIDC authorization.
 - Published events to 3 downstream consumers over Kafka with audit logging.
- **Go, Redis, Docker, Prometheus, GitHub Actions** github.com/BetV3/rate-limiter
Rate Limiter Service
 - Engineered a Go service sustaining 8,000+ req/s at p99 under 5 ms.
 - Implemented 3 rate-limiting algorithms and selected between them through measured trade-offs.
 - Containerized with Docker, instrumented with Prometheus, and tested and built through GitHub Actions CI.
- **Python, PostgreSQL, llama.cpp, Pydantic, Vault, self-hosted GPU** eramirez.dev
Multi-Agent Automation Platform
 - Built and operate an LLM agent orchestration platform: a supervisor delegating to domain managers and ephemeral workers over a durable Postgres-backed queue, so long-running agent work survives restarts and failures.
 - Made non-deterministic output trustworthy structurally: Pydantic-enforced output schemas, verification of every generated value against source-of-truth database rows, and human review gates across a 10-stage pipeline.
 - Runs self-hosted LLM inference with llama.cpp on own GPU hardware; secrets managed with HashiCorp Vault.

Kubernetes, VictoriaMetrics, Grafana, PostgreSQL, Python, Cloudflare, PKI

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- *Platform & Reliability Engineering, self-operated fleet (7 hosts, 40 VMs, 3 Kubernetes clusters)*
 - Operate 3 Kubernetes clusters (15 nodes) across 7 ESXi hosts and 40 VMs; collect 77 scrape targets at 23.6M rows/hr into VictoriaMetrics and run 109 alert signals, each with an owner and a runbook.
 - Found that monitoring steps is not monitoring results: a pipeline reported green for 5 days while producing nothing. Added output-based signals (the first fired on install) and moved the watchdog outside the scheduler it watched after a job failed 970 times in 4 days with one alert.
 - Root-caused a task-queue reaper that could never rescue orphaned tasks (`NULL < now()` is `NULL` in SQL); fixed it with tests in both directions so it neither misses stale tasks nor requeues live ones.
 - Split a queue-depth alert that was red in 369 of 816 runs (45%) into an incident signal and a human-nudge signal, so a page always means the system is broken.
 - Run an internal CA issuing 24-hour certificates with automated renewal for 4 services; caught a silent renewal failure 6h52m before expiry and fixed a renewer restarting 67 times a day.
 - Backups are restore-tested (9,849 rows read back from a restored database); the public edge is Cloudflare Tunnels with 0 inbound ports.

Professional Experience

Walmart

Plano, TX

- *Personal Shopper*
Jul 2023–Present
 - Fulfilled 80+ customer orders per shift with under 1% substitution error rate.
 - Consistently met same-day SLAs in a high-volume environment.

Education

University of Texas at Dallas

Richardson, TX

- *Bachelor of Science in Computer Science*
Graduated August 2025.
Aug 2025

Languages

- English (native).

References

- Available upon request.